

Background Needed to Understand and Implement Algorithm 5.1

in Elsenhans–Jahnel, *Moduli Spaces and the Inverse Galois Problem for Cubic Surfaces*

Abstract

This note reorganizes the material of Sections 1–5 and Appendix A of Elsenhans–Jahnel into the order needed to understand Algorithm 5.1. The emphasis is on definitions, data flow, and the logical role of each construction; proofs are omitted. The central point is that the algorithm never searches directly among cubic equations. Instead, it explicitly descends Coble’s projective model of the moduli space of *marked* cubic surfaces, searches for a rational point on the resulting twist, converts that point from Coble coordinates to Clebsch invariants, and only then reconstructs a cubic equation by a second Galois descent.

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1 The algorithm in one diagram

The input is

$$G \subseteq W(E_6), \quad L/\mathbf{Q} \text{ a finite Galois extension}, \quad \rho : \text{Gal}(L/\mathbf{Q}) \xrightarrow{\sim} G,$$

where the inclusion $G \subseteq W(E_6)$ specifies the desired permutation action on the 27 lines. Algorithm 5.1 follows the chain

$$\begin{aligned}
& \boxed{(G, L, \rho)} \longrightarrow \boxed{V \text{ and } \widetilde{M}_\rho \subset \mathbf{P}(V)} \longrightarrow \boxed{P \in \widetilde{M}_\rho(\mathbf{Q})} \longrightarrow \boxed{(\gamma_1, \dots, \gamma_{40})} \\
& \longrightarrow \boxed{[A : B : C : D : E]} \longrightarrow \boxed{g(T) = \prod_{i=0}^4 (T - a_i)} \longrightarrow \boxed{F_C(X_0, X_1, X_2, X_3) = 0}.
\end{aligned}$$

Here:

- $\widetilde{M}_\rho$ is a twisted form of Coble's compactified moduli space of marked cubic surfaces;
- the γ_i are Coble's 40 irrational invariants;
- $[A : B : C : D : E] \in \mathbf{P}(1, 2, 3, 4, 5)$ is the Clebsch invariant vector of the underlying unmarked cubic surface;

- the roots a_i are the coefficients of a pentahedral presentation; and
- the last arrow is the trace construction of Algorithm A.10.

There are therefore two distinct descents:

1. the descent of the *marked moduli space*, controlled by the input homomorphism ρ and responsible for the prescribed action on the 27 lines; and
2. the descent of a pentahedral equation from a degree-5 étale algebra, used only to turn the already determined Clebsch invariants into an explicit cubic polynomial over $\mathbf{Q}$.

These descents should not be conflated.

2 Cubic surfaces, markings, and the Weyl group

2.1 Smooth cubic surfaces and their lines

A smooth cubic surface over a field K is a smooth degree-3 hypersurface $C \subset \mathbf{P}_K^3$. Over an algebraic closure it contains exactly 27 lines. Their incidence graph is independent of C . The group of permutations of these 27 lines preserving the incidence pairing is the Weyl group

$$W(E_6), \quad |W(E_6)| = 51,840.$$

Equivalently, after identifying a cubic surface with a degree-3 del Pezzo surface, $W(E_6)$ is the group of automorphisms of the geometric Picard lattice preserving the canonical class and the intersection form.

2.2 The blow-up model and a marking

A smooth cubic surface over an algebraically closed field is the anticanonical model of the blow-up of $\mathbf{P}^2$ in six points

$$p_1, \dots, p_6$$

in general position: no three are collinear and the six do not lie on a conic. The 27 lines then have the standard labels

$$l_1, \dots, l_6, \quad l'_1, \dots, l'_6, \quad l''_{ij} \quad (1 \leq i < j \leq 6),$$

where l_i is the exceptional curve over p_i , l'_i is the strict transform of the conic through the five points other than p_i , and l''_{ij} is the strict transform of the line through p_i and p_j .

In the paper, a *marking* is an ordered sequence

$$(l_1, \dots, l_6)$$

of six mutually disjoint lines. Contracting them recovers an ordered six-tuple $(p_1, \dots, p_6)$ in $\mathbf{P}^2$. A marking therefore rigidifies the surface enough to produce a fine moduli space. The set of abstract markings is acted on transitively by $W(E_6)$; changing the marking changes the labels of all 27 lines but not the underlying cubic surface.

2.3 The two moduli spaces

Let $U \subset (\mathbf{P}^2)^6$ be the open locus of ordered six-tuples in general position. The moduli space of marked cubic surfaces is

$$\widetilde{\mathcal{M}} = U / \mathrm{PGL}_3.$$

It is a smooth affine fourfold and is a fine moduli space. The Weyl group acts on it by changing the marking:

$$T_g : \widetilde{\mathcal{M}} \longrightarrow \widetilde{\mathcal{M}}, \quad g \in W(E_6).$$

The quotient

$$\mathcal{M} = \widetilde{\mathcal{M}} / W(E_6)$$

is the coarse moduli space of smooth, unmarked cubic surfaces. The forgetful map is denoted

$$\mathrm{pr} : \widetilde{\mathcal{M}} \longrightarrow \mathcal{M}.$$

Generically, its degree is $|W(E_6)| = 51,840$.

3 Coble's irrational invariants and the gamma variety

3.1 The basic determinants

Write the six points as row vectors

$$p_i = (x_{i,0} : x_{i,1} : x_{i,2}) \in \mathbf{P}^2.$$

For a triple $I = \{i, j, k\}$, let

$$m_I = m_{\{i,j,k\}} := \det \begin{pmatrix} x_{i,0} & x_{i,1} & x_{i,2} \\ x_{j,0} & x_{j,1} & x_{j,2} \\ x_{k,0} & x_{k,1} & x_{k,2} \end{pmatrix}.$$

Thus $m_I = 0$ precisely when the corresponding three points are collinear. Also define

$$d_2 := \det \begin{pmatrix} x_{1,0}^2 & x_{1,1}^2 & x_{1,2}^2 & x_{1,0}x_{1,1} & x_{1,0}x_{1,2} & x_{1,1}x_{1,2} \\ x_{2,0}^2 & x_{2,1}^2 & x_{2,2}^2 & x_{2,0}x_{2,1} & x_{2,0}x_{2,2} & x_{2,1}x_{2,2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{6,0}^2 & x_{6,1}^2 & x_{6,2}^2 & x_{6,0}x_{6,1} & x_{6,0}x_{6,2} & x_{6,1}x_{6,2} \end{pmatrix}.$$

The determinant d_2 vanishes when the six points lie on a conic.

All expressions below have the same multidegree $(3, 3, 3, 3, 3, 3)$ and hence descend to sections of one line bundle $\mathcal{L}$ on $\widetilde{\mathcal{M}}$. Their simultaneous values are therefore projective coordinates.

3.2 The forty invariants

Coble's invariants come in two combinatorial types.

Type 3 + 3. For a partition of $\{1, \dots, 6\}$ into two triples,

$$\gamma_{(i_1 i_2 i_3)(i_4 i_5 i_6)} := m_{\{i_1, i_2, i_3\}} m_{\{i_4, i_5, i_6\}} d_2.$$

The two triples are unordered, so there are $\binom{6}{3}/2 = 10$ such invariants.

Type 2 + 2 + 2. For a cyclically ordered decomposition into three unordered pairs,

$$\begin{aligned} \gamma_{(i_1 i_2)(i_3 i_4)(i_5 i_6)} &:= m_{\{i_1, i_3, i_4\}} m_{\{i_2, i_3, i_4\}} m_{\{i_3, i_5, i_6\}} m_{\{i_4, i_5, i_6\}} \\ &\quad \cdot m_{\{i_5, i_1, i_2\}} m_{\{i_6, i_1, i_2\}}. \end{aligned}$$

Cyclic permutation of the three pairs leaves the symbol unchanged, but reversing their order gives the other orientation. There are 30 invariants of this type.

The historical name “irrational invariants” does not mean that their values are irrational numbers. It refers to their classical role as invariants of a *marked* surface rather than symmetric invariants of the unmarked surface.

3.3 The gamma map and its equations

The 40 sections define Coble’s gamma map

$$\gamma : \widetilde{\mathcal{M}} \hookrightarrow \mathbf{P}^{39}, \quad C^{\text{mk}} \mapsto (\gamma_1 : \dots : \gamma_{40}).$$

Although there are 40 coordinates, they span only a 10-dimensional vector space. Hence the image lies in a linear $\mathbf{P}^9 \subset \mathbf{P}^{39}$. Denote its Zariski closure by

$$\widetilde{M} = \overline{\gamma(\widetilde{\mathcal{M}})}.$$

Theorem 2.7 states that, inside this $\mathbf{P}^9$, $\widetilde{M}$ is cut out by 30 cubic equations. It is a fourfold, so these 30 equations are highly dependent; the variety is not a complete intersection.

A representative cubic relation is

$$\begin{aligned} &\gamma_{(12)(34)(56)} \gamma_{(23)(45)(16)} \gamma_{(14)(36)(25)} \\ &= \gamma_{(12)(36)(45)} \gamma_{(34)(25)(16)} \gamma_{(56)(14)(23)}. \end{aligned}$$

The $W(E_6)$ -orbit of such a relation spans the 30-dimensional space of cubic relations.

A convenient choice of basis for the 10-dimensional span is furnished by the ten invariants of type 3 + 3. Once a basis

$$b_1, \dots, b_{10}$$

has been fixed, every one of the 40 invariants is an explicit rational linear combination of the b_j .

3.4 Why the paper uses eighty signed coordinates

The action of $W(E_6)$ on the 40 coordinates is a *signed* permutation: a group element can send γ_i to $-\gamma_j$. To replace signed permutation matrices by honest permutation matrices, the paper enlarges the coordinate set to

$$\{\pm\gamma_1, \dots, \pm\gamma_{40}\}.$$

There is then a transitive permutation representation

$$\Pi : W(E_6) \hookrightarrow S_{80}.$$

The extra 40 coordinates do not enlarge the underlying 10-dimensional space; the relations $(-\gamma_i) = -\gamma_i$ are part of the resulting 70 linear relations among the 80 coordinates.

This redundancy is computationally useful:

- the full Weyl-group action is represented by coordinate permutations;

- the ten-dimensional basis keeps the linear algebra small; and
- the remaining coordinates are recovered by a fixed 80×10 rational matrix.

More concretely, let

$$B \in M_{80 \times 10}(\mathbf{Q})$$

be the matrix whose rows express the 80 signed coordinates in the chosen basis. If $y \in L^{10}$ is a coordinate vector in the basis $b_1, \dots, b_{10}$, then the corresponding signed gamma vector is

$$x = By \in L^{80}.$$

For $g \in W(E_6)$, let P_g be the 80×80 permutation matrix implementing

$$(T_g x)_i = x_{\Pi(g)^{-1}(i)}.$$

The induced 10×10 matrix R_g is characterized by

$$BR_g = P_g B. \tag{1}$$

This identity is a useful way to eliminate all inverse-versus-transpose ambiguities in an implementation.

Universal precomputation. The following data depend only on Coble's model, not on the eventual group or number field:

1. an ordering of the 80 signed invariants;
2. the permutations $\Pi(g)$ for generators of $W(E_6)$;
3. a choice of ten basis invariants and the matrix B ;
4. a basis of the 30 cubic relations in ten variables; and
5. the formulas converting gamma coordinates to Clebsch invariants.

Algorithm 5.1 assumes that these data have already been made explicit. The paper describes how they arise, but does not print all permutation and relation tables.

4 The unmarked moduli space and Clebsch invariants

4.1 Weighted projective coordinates

Clebsch's invariant theory gives an open embedding

$$\text{Cl} : \mathcal{M} \hookrightarrow \mathbf{P}(1, 2, 3, 4, 5).$$

The homogeneous coordinates are denoted

$$[A : B : C : D : E].$$

Weighted scaling means

$$[A : B : C : D : E] \sim [\lambda A : \lambda^2 B : \lambda^3 C : \lambda^4 D : \lambda^5 E].$$

Thus this vector specifies the geometric isomorphism class of the unmarked cubic surface, away from the usual issues caused by automorphisms over the base field.

4.2 Pentahedral form

A cubic surface with a proper pentahedron admits a Sylvester pentahedral presentation

$$a_0X_0^3 + a_1X_1^3 + a_2X_2^3 + a_3X_3^3 + a_4X_4^3 = 0, \quad X_0 + X_1 + X_2 + X_3 + X_4 = 0. \quad (2)$$

Let σ_i be the elementary symmetric functions in $a_0, \dots, a_4$. Properness is equivalent to

$$\sigma_5 = a_0a_1a_2a_3a_4 \neq 0.$$

On this family the Clebsch invariants are

$$\begin{aligned} A &= \sigma_2^4 - 4\sigma_3\sigma_5, \\ B &= \sigma_1\sigma_5^3, \\ C &= \sigma_4\sigma_5^4, \\ D &= \sigma_2\sigma_5^6, \\ E &= \sigma_5^8. \end{aligned} \quad (3)$$

These have ordinary degrees 8, 16, 24, 32, 40, respectively, which explains the weights 1, 2, 3, 4, 5 after dividing by the common factor 8.

4.3 Recovering the pentahedral symmetric functions

Appendix A.8 inverts [equation \(3\)](#) on the open set $E \neq 0$. After making the weighted scaling prescribed there, one may take

$$[\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5] = \left[B, D, \frac{C^2 - AE}{4}, CE, E^2 \right]. \quad (4)$$

The five pentahedral coefficients a_i are then the roots of

$$g(T) = T^5 - \sigma_1T^4 + \sigma_2T^3 - \sigma_3T^2 + \sigma_4T - \sigma_5. \quad (5)$$

This quintic is an auxiliary object. Its splitting field and Galois group are generally unrelated to the input extension $L/\mathbf{Q}$ and the prescribed subgroup $G \subseteq W(E_6)$.

5 The explicit map from Coble coordinates to Clebsch invariants

Theorem 3.9 is the bridge from the marked gamma model to the unmarked Clebsch model. Choose one coordinate from each pair $\{\gamma_i, -\gamma_i\}$ and set

$$P_k := \sum_{i=1}^{40} \gamma_i^k.$$

Only even values of k occur, so the choice of signs is irrelevant. The formulas are

$$\begin{aligned}
A &= -6P_2, \\
B &= -24P_4 + \frac{41}{16}P_2^2, \\
C &= \frac{576}{13}P_6 - \frac{396}{13}P_4P_2 + \frac{29}{13}P_2^3, \\
D &= -\frac{62208}{1171}P_8 + \frac{54864}{1171}P_6P_2 + \frac{203616}{1171}P_4^2 \\
&\quad - \frac{61287}{1171}P_4P_2^2 + \frac{13393}{4684}P_2^4, \\
E &= \frac{41472}{155}P_{10} - \frac{4605984}{36301}P_8P_2 - \frac{106272}{403}P_6P_4 \\
&\quad + \frac{19990440}{471913}P_6P_2^2 + \frac{47719206}{471913}P_4^2P_2 \\
&\quad - \frac{7468023}{471913}P_4P_2^3 + \frac{10108327}{18876520}P_2^5.
\end{aligned} \tag{6}$$

Each right-hand side has degree 2, 4, 6, 8, 10, respectively, in the gamma coordinates. Therefore scaling all gamma coordinates by μ changes (A, B, C, D, E) by

$$(\mu^2 A, \mu^4 B, \mu^6 C, \mu^8 D, \mu^{10} E),$$

which is precisely weighted scaling with parameter $\lambda = \mu^2$. Thus the map is well-defined on projective gamma coordinates.

Important factor-of-two warning. Formula [equation \(6\)](#) uses the 40 original invariants. If an implementation sums over all 80 signed coordinates, then for even k one gets

$$\sum_{j=1}^{80} x_j^k = 2P_k.$$

The formulas must be adjusted accordingly, or one must select exactly one representative from each signed pair.

6 Twisting the marked moduli space

6.1 The prescribed Galois action

Let K be a field of characteristic zero and fix a continuous homomorphism

$$\rho : \text{Gal}(\overline{K}/K) \longrightarrow W(E_6).$$

The twisted moduli problem asks for marked cubic surfaces over $\overline{K}$ whose labeled lines transform under Galois according to ρ . Theorem 4.2 states that this functor is represented by a K -scheme $\widetilde{\mathcal{M}}_\rho$, a twist of $\widetilde{\mathcal{M}}$.

For Algorithm 5.1, $K = \mathbf{Q}$, and ρ factors as

$$\text{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \twoheadrightarrow \text{Gal}(L/\mathbf{Q}) \xrightarrow{\sim} G \hookrightarrow W(E_6).$$

Over L , the twist becomes the ordinary marked moduli space:

$$\widetilde{\mathcal{M}}_\rho \times_{\mathbf{Q}} L \cong \widetilde{\mathcal{M}}_L.$$

6.2 The descent datum

For $\sigma \in \text{Gal}(L/\mathbf{Q})$, there are two operations on $\widetilde{\mathcal{M}}_L$:

1. σ acts on the coefficients of the L -scheme; and
2. $T_{\rho(\sigma)}$ relabels the 27 lines.

The twist is obtained from the semilinear descent datum

$$d_\sigma := T_{\rho(\sigma)} \circ \sigma.$$

A $\mathbf{Q}$ -rational point of $\widetilde{\mathcal{M}}_\rho$ corresponds to an L -point of $\widetilde{\mathcal{M}}_L$ fixed by this datum. After forgetting the marking, this gives a cubic surface over $\mathbf{Q}$ for which Galois acts on the 27 lines through the prescribed homomorphism ρ .

This is the conceptual heart of the inverse Galois construction: one does not impose equations on the lines themselves. One changes the rational structure on the moduli space of *marked* surfaces so that a rational point is exactly a surface with the desired nontrivial action on its labels.

6.3 The coordinate fixed-point condition

In the 80 signed gamma coordinates, the relabeling operation is

$$T_{\rho(\sigma)}(x_0, \dots, x_{79}) = (x_{\Pi(\rho(\sigma))^{-1}(0)}, \dots, x_{\Pi(\rho(\sigma))^{-1}(79)}).$$

Thus the affine-cone fixed-point condition is

$$(\sigma(x_{\Pi(\rho(\sigma))^{-1}(0)}), \dots, \sigma(x_{\Pi(\rho(\sigma))^{-1}(79)})) = (x_0, \dots, x_{79}). \quad (7)$$

Projectively, equality is only required up to a common scalar. The paper uses the stronger affine equality because the fixed vectors form an actual $\mathbf{Q}$ -vector space; projectivizing that descended vector space gives the desired descended ambient projective space. This is the linear form of Hilbert–90 behind Remarks 4.4.

In the ten-dimensional basis, let $g = \rho(\sigma)$ and use the matrix R_g from [equation \(1\)](#). Since B has rational entries, condition [equation \(7\)](#) is equivalent to

$$R_g \sigma(y) = y, \quad y \in L^{10}. \quad (8)$$

The action is semilinear: R_g mixes the ten gamma coordinates, while σ acts on each coefficient in L .

6.4 Turning semilinear equations into rational linear algebra

Choose a $\mathbf{Q}$ -basis $e_1, \dots, e_n$ of L , where $n = [L : \mathbf{Q}]$. For each chosen generator g of G , let

$$S_g \in \text{GL}_n(\mathbf{Q})$$

be the matrix of the field automorphism $\sigma_g := \rho^{-1}(g)$ in this basis. Expanding

$$y_j = \sum_{r=1}^n Y_{jr} e_r$$

turns [equation \(8\)](#) into a homogeneous $\mathbf{Q}$ -linear system in $10n$ unknowns. With the convention that the n field coordinates of each y_j are consecutive, the matrix for the generator g is

$$R_g \otimes S_g - I_{10n}. \quad (9)$$

The paper writes the system in the 80 redundant coordinates, producing $80n$ equations per generator, but the unknown count remains $10n$ after the linear relations are imposed.

The common solution space is

$$V := \{y \in L^{10} : R_g \sigma_g(y) = y \text{ for every } g \in G\}.$$

Galois descent guarantees

$$\dim_{\mathbf{Q}} V = 10.$$

This dimension is an important implementation check. A different dimension usually signals an inconsistent convention for $\Pi(g)$, P_g , R_g , or the direction of the isomorphism ρ .

6.5 Descending the cubic equations

Let $v_1, \dots, v_{10}$ be a $\mathbf{Q}$ -basis of V , with each $v_i \in L^{10}$. Introduce rational projective coordinates $z_1, \dots, z_{10}$ and substitute

$$y = z_1 v_1 + \dots + z_{10} v_{10}$$

into the 30 universal cubic equations of $\widetilde{M}$. The resulting coefficients lie in $\mathbf{Q}$ because the relation space is stable under the descent datum. The output is

$$F_1(z) = \dots = F_{30}(z) = 0, \quad F_i \in \mathbf{Q}[z_1, \dots, z_{10}],$$

defining the descended compactification

$$\widetilde{M}_\rho \subset \mathbf{P}(V) \cong \mathbf{P}_{\mathbf{Q}}^9.$$

An L -valued gamma vector is recovered from a rational coordinate vector $z \in \mathbf{Q}^{10}$ by first forming $y = \sum z_i v_i$ and then $x = By$.

7 Algorithm 5.1, expanded step by step

Input data

One needs more than an abstract isomorphism type of a finite group. The input consists of:

1. a concrete subgroup $G \subseteq W(E_6)$, specified as permutations of the 27 line labels or through compatible Weyl-group generators;
2. a finite Galois number field $L/\mathbf{Q}$ with $\text{Gal}(L/\mathbf{Q}) \cong G$;
3. an explicit isomorphism $\rho : \text{Gal}(L/\mathbf{Q}) \xrightarrow{\sim} G$; and
4. exact arithmetic in L , including explicit matrices for the relevant field automorphisms.

Conjugate subgroups of $W(E_6)$ represent the same line action up to relabeling, but nonconjugate embeddings of an isomorphic abstract group represent different actions.

7.1 Step (i): fix the Weyl-group coordinate action

Choose generators $\Gamma \subseteq G$. For every $g \in \Gamma$, store

$$\Pi(g) \in S_{80}.$$

Fix once and for all ten linearly independent signed invariants and the matrix B expressing the other seventy in terms of them.

This step is universal group-theoretic and invariant-theoretic bookkeeping. It turns the geometric operation “change the marking by g ” into a concrete permutation of coordinates.

7.2 Step (ii): compress from eighty coordinates to ten

For each $g \in \Gamma$, compute the 10×10 matrix R_g by solving

$$BR_g = P_g B.$$

This is the representation of g on the ten-dimensional gamma coordinate space in the chosen basis.

The 80-coordinate representation is sparse and transparent; the 10-coordinate representation is the one used for efficient linear algebra.

7.3 Step (iii): make the field action explicit

Choose a $\mathbf{Q}$ -basis of L . For each $g \in \Gamma$, compute the matrix S_g of

$$\sigma_g = \rho^{-1}(g) \in \text{Gal}(L/\mathbf{Q})$$

as a $\mathbf{Q}$ -linear map $L \rightarrow L$.

This is the arithmetic half of the semilinear action. Step (ii) describes how g permutes moduli coordinates; Step (iii) describes how the corresponding Galois automorphism changes their coefficients.

7.4 Step (iv): compute the descended ten-dimensional space

Solve

$$R_g \sigma_g(y) = y \quad (g \in \Gamma)$$

as a rational linear system. Equivalently, intersect the kernels of the matrices [equation \(9\)](#). The result is a ten-dimensional $\mathbf{Q}$ -space

$$V \subset L^{10}.$$

The paper's implementation refines this step arithmetically. Instead of solving only in L^{10} , it intersects with $\mathcal{O}_L^{10}$, where $\mathcal{O}_L$ is the maximal order. This produces a rank-10 $\mathbf{Z}$ -lattice. Under the Minkowski embedding it becomes a Euclidean lattice, and LLL reduction yields a basis $v_1, \dots, v_{10}$ with comparatively small algebraic coefficients. Using this LLL-reduced basis as projective coordinates greatly improves the subsequent small-height rational-point search.

7.5 Step (v): descend the thirty cubics

Substitute

$$y = z_1 v_1 + \dots + z_{10} v_{10}$$

into the 30 universal cubic equations of Coble's gamma variety. Simplify the coefficients and verify that they lie in $\mathbf{Q}$. This gives the explicit twisted variety

$$\widetilde{M}_\rho = V(F_1, \dots, F_{30}) \subset \mathbf{P}_{\mathbf{Q}}^9.$$

At this point the desired Galois behavior has been encoded entirely in rational equations. The extension L remains visible only through the coefficients of the embedding of V into the original gamma coordinate space.

7.6 Step (vi): search for a rational point

Search for

$$P = [z_1 : \cdots : z_{10}] \in \widetilde{M}_\rho(\mathbf{Q}).$$

The authors use a naive bounded-height search; in primitive integral coordinates this is essentially a search through integer vectors satisfying all 30 cubics. The LLL basis from Step (iv) is crucial because it often puts a useful rational point at very small height.

The equations define the compactification $\widetilde{M}_\rho$, not only the open moduli space $\widetilde{\mathcal{M}}_\rho$. A candidate must therefore be checked to lie in the open locus corresponding to a smooth marked cubic surface. In practice, the reconstruction and smoothness checks in Step (vii) reject many boundary or exceptional points.

7.7 Step (vii): recover a cubic equation

Starting from z , perform the following sequence.

(a) **Recover gamma coordinates.** Compute

$$y = \sum_{i=1}^{10} z_i v_i \in L^{10}, \quad x = By \in L^{80}.$$

Select one coordinate from each signed pair to obtain $\gamma_1, \dots, \gamma_{40}$.

Although these values generally lie in L , their even power sums are rational. Indeed, Galois sends the gamma vector to a signed permutation of itself, so P_{2k} is fixed by $\text{Gal}(L/\mathbf{Q})$.

(b) **Compute Clebsch invariants.** Evaluate $P_2, P_4, P_6, P_8, P_{10}$ and use [equation \(6\)](#) to obtain

$$[A : B : C : D : E] \in \mathbf{P}(1, 2, 3, 4, 5)(\mathbf{Q}).$$

This forgets the marking and retains the isomorphism class of the underlying cubic surface.

(c) **Recover pentahedral symmetric data.** If $E \neq 0$, use [equation \(4\)](#) and form the quintic [equation \(5\)](#). Its roots are the five pentahedral coefficients over a splitting field.

(d) **Descend the pentahedral equation.** Apply Algorithm A.10, described in the next section, to produce a homogeneous cubic polynomial over $\mathbf{Q}$.

(e) **Validate.** Check that the quintic is separable, that the resulting cubic surface is smooth, and that recomputed invariants agree with the original weighted Clebsch point. If the equation problem fails, return to Step (vi) and choose a different rational point.

8 Algorithm A.10 in conceptual form

Let K be a characteristic-zero field and suppose the quintic

$$g(T) = T^5 - \sigma_1 T^4 + \sigma_2 T^3 - \sigma_3 T^2 + \sigma_4 T - \sigma_5$$

is separable. Set

$$A_g := K[T]/(g), \quad \theta := T \bmod g.$$

This is a degree-5 étale K -algebra. It may be a field or a product of fields; no factorization or splitting field is required.

8.1 The trace-zero space

Compute

$$t_j := \text{Tr}_{A_g/K}(\theta^j), \quad j = 0, \dots, 4.$$

The trace-zero subspace

$$N := \ker(\text{Tr}_{A_g/K} : A_g \rightarrow K)$$

has dimension 4. Relative to the basis $1, \theta, \dots, \theta^4$, it is the kernel of the row matrix

$$(t_0 \ t_1 \ t_2 \ t_3 \ t_4).$$

Choose a basis C_0, C_1, C_2, C_3 of N , writing

$$C_i = \sum_{j=0}^4 c_i^{(j)} \theta^j.$$

8.2 The trace cubic

Form the generic trace-zero element

$$u(X) := C_0 X_0 + C_1 X_1 + C_2 X_2 + C_3 X_3 \in A_g[X_0, X_1, X_2, X_3].$$

Algorithm A.10 outputs

$$F_C(X_0, X_1, X_2, X_3) := \text{Tr}_{A_g/K}(\theta u(X)^3). \quad (10)$$

Computationally, one reduces θu^3 modulo $g(T)$ and applies the trace to each coefficient. The result is a homogeneous cubic in four variables with 20 coefficients in K .

To see the geometry encoded by [equation \(10\)](#), extend scalars to a splitting field of g . Then

$$A_g \otimes_K \overline{K} \cong \overline{K}^5,$$

and θ maps to $(a_0, \dots, a_4)$. The element u maps to five linear forms $\ell_0, \dots, \ell_4$. Because u has trace zero,

$$\ell_0 + \dots + \ell_4 = 0,$$

and the trace cubic becomes

$$a_0 \ell_0^3 + \dots + a_4 \ell_4^3.$$

Thus it is exactly the pentahedral surface, written in four K -rational coordinates.

9 The complete logical chain

The correctness logic of Algorithm 5.1 can be read as the following sequence, with no proof details needed for implementation.

1. A marking identifies the abstract configuration of 27 lines with the standard labeled configuration on which $W(E_6)$ acts.
2. The homomorphism $\rho : \text{Gal}(L/\mathbf{Q}) \rightarrow G \subseteq W(E_6)$ prescribes how Galois should change this labeling.
3. The twist $\widetilde{\mathcal{M}}_\rho$ is defined so that its $\mathbf{Q}$ -points are exactly marked surfaces over L whose Galois conjugates are identified by the corresponding relabelings.

4. Coble’s gamma embedding linearizes the Weyl-group action. The twisted Galois condition therefore becomes the semilinear linear system $R_g \sigma_g(y) = y$.
5. The fixed vectors form the descended ten-dimensional $\mathbf{Q}$ -space V , and the 30 Weyl-stable cubic relations descend to equations for $\widetilde{M}_\rho \subset \mathbf{P}(V)$.
6. A rational point on the open part of $\widetilde{M}_\rho$ yields a cubic surface over $\mathbf{Q}$ whose 27 lines carry the prescribed Galois action.
7. The rational point gives gamma coordinates in L . Weyl-invariant even power sums give a Clebsch point over $\mathbf{Q}$, thereby forgetting the marking.
8. On the open set $E \neq 0$, the Clebsch point determines the symmetric functions of five pentahedral coefficients and hence a separable quintic $g(T)$.
9. The trace construction in the étale algebra $\mathbf{Q}[T]/(g)$ produces a cubic equation over $\mathbf{Q}$ geometrically representing the same Clebsch point.
10. Away from the automorphism and discriminant divisors, this equation is a smooth cubic surface with exactly the desired action on its lines.

10 Failure modes and validation checks

10.1 No rational point on the chosen twist

The point search may find no rational point. This does not show that the abstract subgroup cannot occur. A different Galois extension $L/\mathbf{Q}$ with the same group, or a different explicit identification with G , produces a different Galois cocycle and can produce a twist with rational points.

10.2 $E = 0$: no proper pentahedron

Since $E = \sigma_5^8$, the condition $E = 0$ means that the surface is outside the open pentahedral locus used by Appendix A.8. The Clebsch point is still meaningful, but this particular equation-solving procedure cannot be applied directly.

10.3 The quintic is inseparable

If g has repeated roots, some pentahedral coefficients coincide. The paper identifies this with the Eckardt-point/automorphism divisor, denoted by the vanishing of a classical invariant F . On this locus Clebsch invariants need not determine the K -isomorphism class uniquely because nontrivial twists can occur. Algorithm A.10 therefore requires g to be separable.

10.4 The resulting cubic is singular

Even when $E \neq 0$ and g is separable, the cubic discriminant Δ may vanish. The output must be checked for smoothness, for example by verifying that the four partial derivatives have no common projective zero over an algebraic closure or by computing the cubic discriminant.

10.5 A point on the boundary of the gamma compactification

The 30 cubics define $\widetilde{M}_\rho$, the compactification. A rational point can lie outside $\widetilde{M}_\rho$. Necessary practical checks include verifying that the recovered gamma vector is nonzero and lies in the expected open locus, that $E \neq 0$, that g is separable, and that the final cubic is smooth.

10.6 Linear-algebra sanity checks

A robust implementation should verify:

- the 80 signed coordinates have rank 10;
- $BR_g = P_g B$ for every generator;
- the matrices R_g satisfy the group relations;
- the semilinear fixed space has $\mathbf{Q}$ -dimension 10;
- substitution into the 30 cubics gives coefficients fixed by $\text{Gal}(L/\mathbf{Q})$, hence rational;
- the even power sums and Clebsch invariants are rational; and
- the Clebsch invariants of the final cubic agree up to weighted scaling with the input Clebsch vector.

11 Three Galois actions that must be kept separate

Action	What it acts on	Its role
$G \subseteq W(E_6)$	The labels of the 27 lines, and hence the signed gamma coordinates	This is the desired geometric Galois action.
$\text{Gal}(L/\mathbf{Q})$ on L	Coefficients of gamma-coordinate vectors in the number field L	This is combined semilinearly with the Weyl action to construct the twist.
The Galois action on the roots of $g(T)$	The five pentahedral coefficients $a_0, \dots, a_4$	This is only an auxiliary descent used to recover a cubic equation from Clebsch invariants. Its field and group need not equal L and G .

12 Implementation-oriented data checklist

Data precomputed once

- a fixed ordering of the 40 Coble invariants and of the 80 signed invariants;
- the signed-pair involution on the 80 indices;
- permutations Π for generators of $W(E_6)$;
- an 80×10 matrix B expressing all signed coordinates in a chosen basis;

- 30 homogeneous cubic forms defining $\widetilde{M} \subset \mathbf{P}^9$;
- the formulas [equation \(6\)](#); and
- routines implementing [equation \(4\)](#) and [equation \(10\)](#).

Data supplied for each run

- generators of the concrete subgroup $G \subseteq W(E_6)$;
- a defining representation of the Galois field $L/\mathbf{Q}$;
- a $\mathbf{Q}$ -basis or integral basis of L ;
- explicit automorphisms of L corresponding to the generators of G ;
- matrices S_g in the chosen field basis; and
- bounds and normalization conventions for the rational-point search.

Main intermediate objects

- the ten-dimensional matrices R_g ;
- the fixed $\mathbf{Q}$ -space $V \subset L^{10}$ and preferably an LLL-reduced integral lattice basis;
- the 30 descended cubic equations over $\mathbf{Q}$;
- a rational point $z \in \mathbf{P}(V)(\mathbf{Q})$;
- the corresponding gamma vector $x \in L^{80}$;
- the Clebsch vector $[A : B : C : D : E]$;
- the pentahedral quintic $g(T)$; and
- the final cubic form in four variables.

13 What Algorithm 5.1 is really doing

The algorithm is best viewed as an invariant-theoretic substitute for an explicit universal family.

- It does *not* begin with a cubic equation and solve for its 27 lines.
- It does *not* descend a known cubic equation over L .
- It first descends a projective model of the moduli space on which the Weyl-group symmetry is linear.
- A rational point on that descended moduli space supplies the desired surface abstractly, together with the prescribed line action.
- Coble-to-Clebsch formulas forget the marking while retaining enough invariant data to identify the geometric cubic surface.
- The pentahedral trace construction then converts that invariant point into a concrete polynomial equation over $\mathbf{Q}$.

This separation of tasks is the main architectural idea:

prescribed line action $\xrightarrow{\text{twisted marked moduli}}$ rational moduli point
rational moduli point $\xrightarrow{\text{invariants and trace}}$ explicit cubic equation.

References

- [1] A.-S. Elsenhans and J. Jahnel, *Moduli spaces and the inverse Galois problem for cubic surfaces*, Trans. Amer. Math. Soc. **367** (2015), no. 11, 7837–7861; [arXiv:1209.5591](#).